

Deterministic Optimization

Geometric Aspects of Linear Optimization

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Extreme Rays and Conic Hull

Geometric Aspects of Linear Optimization

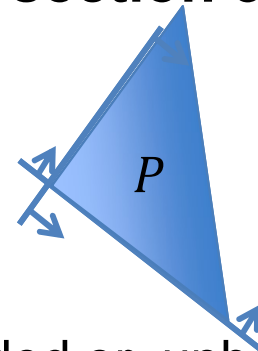
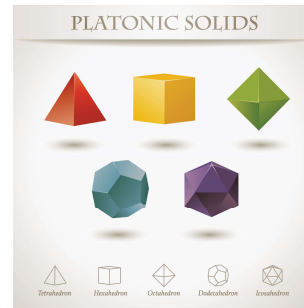
Learning Objectives for this lesson

- Learn another key geometric construction:
 - Conic combination
 - Conic hull
 - Extreme rays

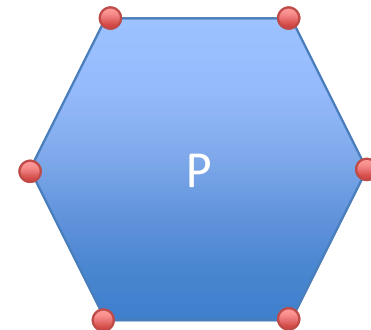
Polyhedron Representations

- Halfspace representation: A **polyhedron** is the **intersection** of a **finite number of halfspaces**.

$$P = \{x \in R^n \mid a_1^\top x \leq b_1, \\ \dots \\ a_m^\top x \leq b_m\}$$

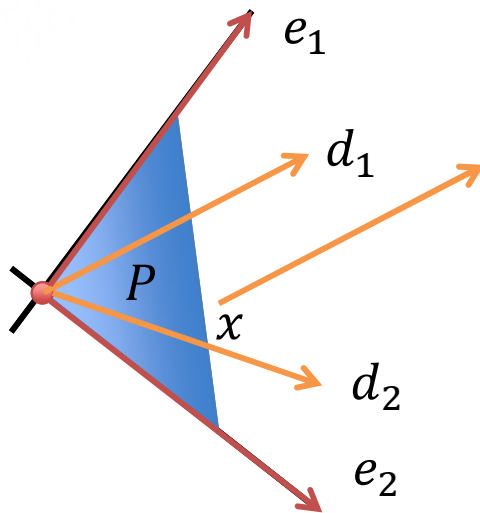


- This representation applies to all polyhedra, bounded or unbounded
- Extreme-point representation: A **bounded** polyhedron is the convex hull of all its extreme points.
 - Convex hull can only generate a bounded set.
 - How to extend extreme-point representation to an **unbounded** polyhedron?



Characterization of Unbounded Polyhedron: Recession Directions

- A polyhedron is unbounded iff there are directions to move to infinity without leaving the polyhedron.

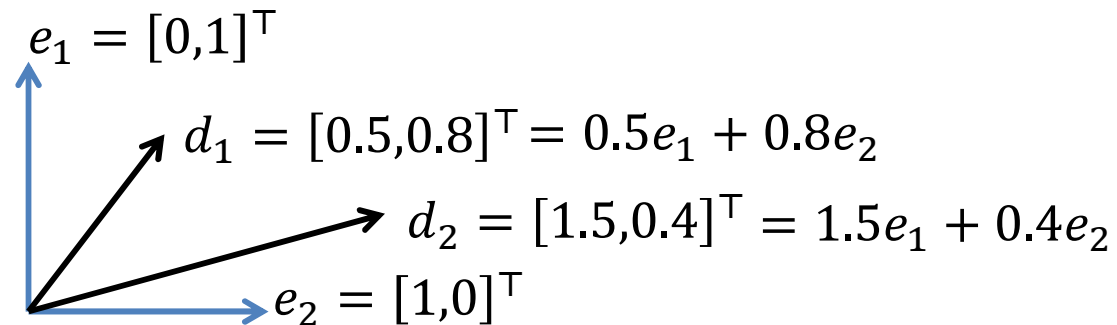


- d_1 is a direction such that starting from any $x \in P$ $x + \theta d_1$ is always in P , for any $\theta \geq 0$
- d_2 is also such a direction, so are e_1, e_2
- d_1, d_2, e_1, e_2 are called **recession directions**
- Recession directions are **rays** that we learned earlier.

- But what is the difference between d_1, d_2 and e_1, e_2 ?
 - e_1, e_2 look somewhat “extreme”. We will quantify this.

Conic Combination

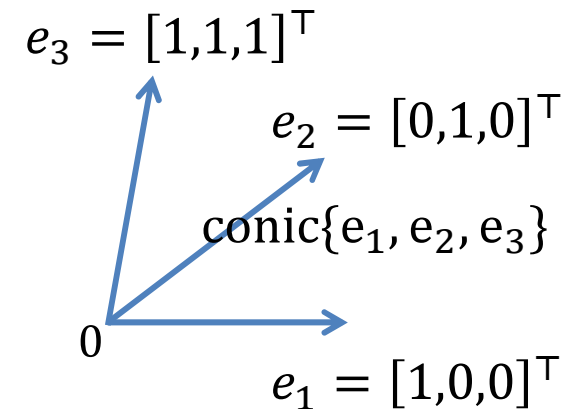
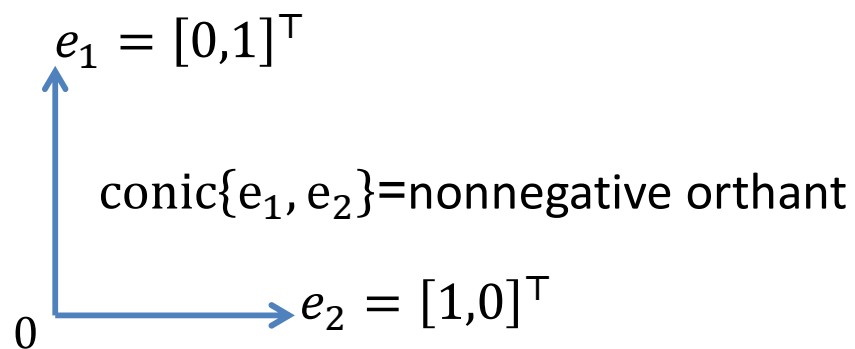
- A ray d is a **conic combination** of two rays e_1, e_2 if d is a **nonnegative** weighted sum of e_1, e_2 .
 - $d = c_1 e_1 + c_2 e_2$ for $c_1, c_2 \geq 0$.
 - Examples:



Conic Hull

- The set of all conic combinations of rays $r_1, \dots, r_m$ is called the **conic hull** of $r_1, \dots, r_m$:

- $\text{conic}\{a_1, \dots, a_n\} = \{x \mid x = \lambda_1 a_1 + \dots + \lambda_n a_n, \lambda_i \geq 0 \ \forall i = 1, \dots, n\}$

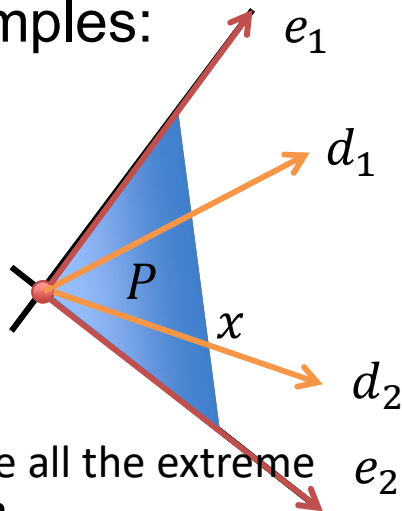


- Such set (i.e. the conic hull) is called a **cone**.

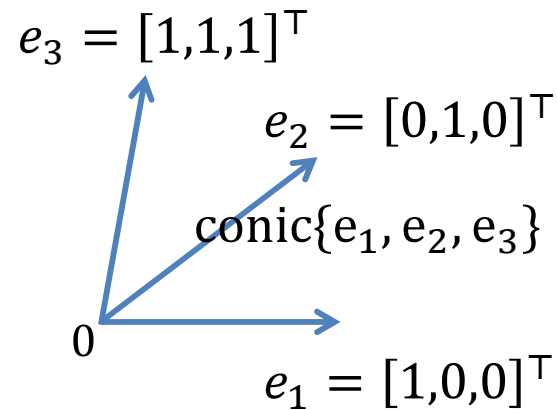
Extreme Rays

- A ray e in a cone C is called an extreme ray if e is not a conic combination of other two different rays in the cone C .
 - Notice the analogy to the definition of an extreme point.

- Examples:



e_1, e_2 are all the extreme rays of P



e_1, e_2, e_3 are all the extreme rays of P

Extreme Ray and Extreme Point

- If a polyhedron is bounded, there is no extreme ray
- If a polyhedron is bounded, there must be an extreme point
- If a polyhedron is unbounded and it does not contain a line, it must have an extreme ray
- If a polyhedron is unbounded, it may not have an extreme point

Summary

- We learned:
 - Conic combination
 - Conic hull
 - Cone
 - Extreme rays
 - Unbounded polyhedron